TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

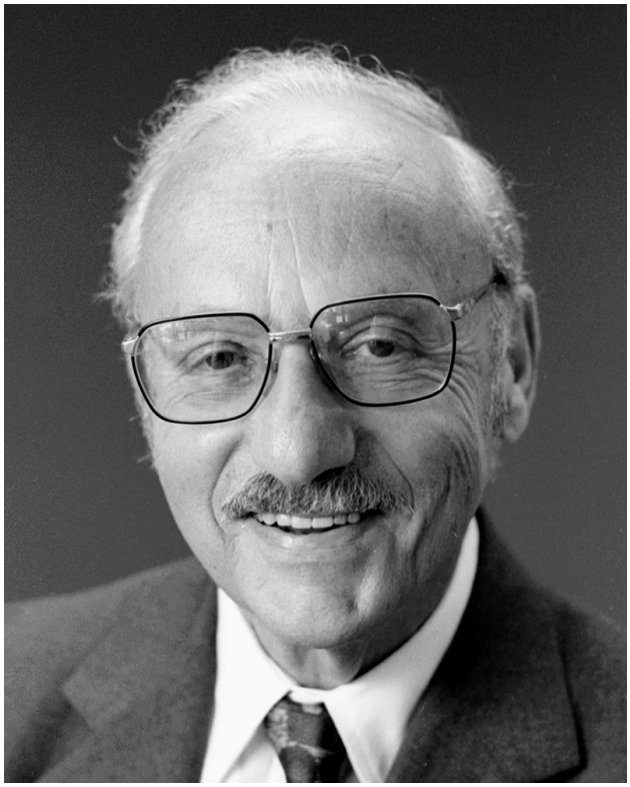


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

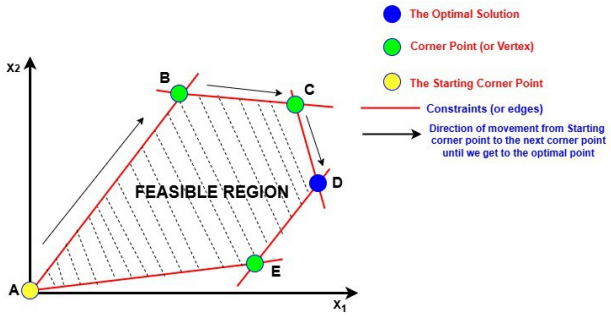


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Problema 1
- Amount of decision variables: 2
- Amount of restrictions: 3
- Total Amount of variables (decision + slack): 5
- Type of problem: Maximization

Equations

$$
\begin{aligned}
&\text{Max } Z = 4\,x_1 + x_2 \\
&\text{subject to:} \\
&\qquad 2\,x_1 + 3\,x_2 \leq 4 \\
&\qquad x_1 + x_2 \leq 1 \\
&\qquad 4\,x_1 + x_2 \leq 2 \\
&\qquad x_1, x_2 \geq 0
\end{aligned}
$$

Initial Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	-4.0000	-1.0000	0.0000	0.0000	0.0000	0.0000
0.0000	2.0000	3.0000	1.0000	0.0000	0.0000	4.0000
0.0000	1.0000	1.0000	0.0000	1.0000	0.0000	1.0000
0.0000	4.0000	1.0000	0.0000	0.0000	1.0000	2.0000

Intermediate Tables

Iteration 1

Entering variable: x_1 (most negative coefficient in the Z row: -4.0000). Leaving variable: s_3 (row 3, minimum ratio 0.5000).

Finding the Pivot

Z	x_1	x_2	s_1	s_2	s_3	b	P
1.0000	-4.0000	-1.0000	0.0000	0.0000	0.0000	0.0000	-
0.0000	2.0000	3.0000	1.0000	0.0000	0.0000	4.0000	2.0000
0.0000	1.0000	1.0000	0.0000	1.0000	0.0000	1.0000	1.0000
0.0000	**4.0000**	1.0000	0.0000	0.0000	1.0000	2.0000	0.5000

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000
0.0000	0.0000	2.5000	1.0000	0.0000	-0.5000	3.0000
0.0000	0.0000	0.7500	0.0000	1.0000	-0.2500	0.5000
0.0000	1.0000	0.2500	0.0000	0.0000	0.2500	0.5000

Final Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000
0.0000	0.0000	2.5000	1.0000	0.0000	-0.5000	3.0000
0.0000	0.0000	0.7500	0.0000	1.0000	-0.2500	0.5000
0.0000	1.0000	0.2500	0.0000	0.0000	0.2500	0.5000

Solution

Solution

The optimal solution is:

- $x_1 = 0.5000$
- $x_2 = 0.0000$ (non-basic)
- $s_1 = 3.0000$
- $s_2 = 0.5000$
- $s_3 = 0.0000$ (non-basic)

The optimal value of the objective function is: $Z = 2.0000$

Multiple solutions

In the final table the non-basic variable x_2 has a coefficient of 0 in the Z row. Entering it into the basis does not change the value of Z, so there is more than one optimal solution.

One more pivot (column x_2, row 2) gives a second final table:

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000
0.0000	0.0000	0.0000	1.0000	-3.3333	0.3333	1.3333
0.0000	0.0000	1.0000	0.0000	1.3333	-0.3333	0.6667
0.0000	1.0000	0.0000	0.0000	-0.3333	0.3333	0.3333

Solution 1: $(0.5000, 0.0000)$
Solution 2: $(0.3333, 0.6667)$

Any convex combination of both solutions is also optimal:

$$\mathbf{x} = \lambda\,\mathbf{x}_1 + (1-\lambda)\,\mathbf{x}_2, \qquad 0 \le \lambda \le 1.$$

Some additional optimal solutions:

- $\lambda = 0.25$: $(0.3750, 0.5000)$
- $\lambda = 0.50$: $(0.4167, 0.3333)$
- $\lambda = 0.75$: $(0.4583, 0.1667)$